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Trading rules, competition for order flow and market fragmentation ☆

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ABSTRACT

We investigate competition between traditional stock exchanges and new dark trading venues using an important difference in regulatory treatment. Securities and Exchange Commission required minimum pricing increments constrain some stock spreads, causing large limit order queues. Dark pools allow some traders to bypass existing limit order queues with minimal price improvement. Using a regression discontinuity design, we find that spread constraints significantly weaken exchanges' competitiveness. As more orders migrate to dark pools, the probability of subsequent order execution there increases, raising liquidity. The ability to circumvent time priority of displayed limit orders is one cause of the rapid rise in US equity market fragmentation.

1. Introduction

Competition between trading venues for order flow is changing the structure of financial markets. Dark trading venues are introducing new trading systems with potentially faster execution, greater anonymity, lower costs of trading, and more liquidity. By early 2014, more than a third of all US stock trading volume takes place on dark trading venues. This reflects the rapid growth in importance of off-exchange trading systems for US equity trades following adoption of Regulation National Market System (Reg NMS), promulgated by the Securities and Exchange Commission (SEC) in 2005. Much of this growth is concentrated in dark pools, which doubled in market share from about 7% in early 2008 to an average of 14.54% in 2013, according to Rosenblatt Securities. Given this rapid growth of competing trading systems, which comes at the expense of traditional stock exchanges,[1] understanding more clearly the competitive advantages of both dark venues and traditional exchanges is important.

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[1] The 2012 NYSE Euronext takeover by Intercontinental Exchange highlights this sea change in share trading.

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In this study, we investigate the effects of the minimum pricing increment regulation on trading venue competition. Imposing on exchanges a minimum pricing increment, or tick size, which is currently one penny, creates artificial constraints on the quoted bid-ask spread. When the spread is constrained, additional trading interest is reflected in a buildup of depth in the exchange's limit order book, which results in long queues at the best bid and ask prices, with time priority on each trading venue for the earliest limit order entering the book. We explore how traders can use dark pools to trade ahead of exchange displayed limit orders, while providing little or no price improvement.[2] As traders begin to migrate their order flow to dark venues, the probability of trade execution there rises, which in turn attracts more trades to dark pools. This positive feedback loop in trading, initially triggered by minimum pricing requirements, is a potentially important cause of the rapid growth in dark pools.

We use regression discontinuity analysis as one method to isolate the marginal effects of the minimum pricing increment regulation on intermarket competition. Specifically, we examine whether tick size constraints give dark venues a competitive advantage over traditional stock exchanges by enabling faster executions at lower cost. For this investigation, unlike previous studies, we use a data set of off-exchange trading classified into five dark venues: dark electronic communication networks (DARK-ECN), block crossing networks (BLOCK), ping destinations (PING), retail market makers (INTERNALIZE), and others (OTHER).[3] Our primary focus is on trading activity on DARK-ECNs relative to traditional exchanges.

The SEC adoption of Reg NMS Rule 612 in 2005 offers an informative natural experiment. Rule 612 prohibits displaying, ranking, or accepting orders priced at more than two decimal places for stocks priced at or above \$1.00 by broker-dealers and exchanges. When stock prices fall below \$1.00, the required minimum pricing increment for exchange trades decreases from a penny, or \$0.01, to \$0.0001. Thus, there is a discrete change in the minimum spread for orders on exchanges when stock prices fluctuate around \$1.00.

While all trading venues must abide by the display, rank, and acceptance conditions of Rule 612, broker-dealers operating in alternative trading systems (ATSs) can offer price improvement in subpenny increments for all stocks regardless of price, which results in trades executing within a penny increment and potentially executing more quickly.[4] When spreads are constrained, long queues of orders at the best bids and offers are likely, creating strong incentives for some traders to shift their orders to alternative trading venues where they can trade immediately at or within the National Best Bid and Offer (NBBO) (see Buti, Rindi, Wen, and Werner, 2011). Thus, when stocks trade at prices above one dollar and exchange spreads are constrained, ATSs have a competitive advantage, which disappears at prices below one dollar, when the SEC-mandated minimum pricing increment for exchanges abruptly falls to one-hundredth of a penny. We propose that many traders take advantage of the faster executions and smaller spreads often available in dark pools to trade at or within the NBBO. Our initial empirical analysis uses regression discontinuity analysis around the \$1.00 price threshold to test this dark pool competitive advantage.

The regression discontinuity design approach is widely used in economics and is becoming more common in corporate finance.[5] The central idea behind the discontinuity design method is that observations just below a cutoff point can be directly compared with observations just above the cutoff. In our application, we compare a dark venue's market share of trading in a stock priced just below \$1.00 with trading in the same stock priced just above \$1.00. By selecting events in which stocks cross the \$1.00 threshold, we can more accurately adjust for stock specific factors because each stock acts as its own control.

Our results indicate that the SEC's Minimum Pricing Increment Rule gives DARK-ECNs a competitive advantage that has persisted over time and provides one explanation for the rapid rise in dark pool market share. We find a discontinuity in the DARK-ECNs' market share of trades as the stock price rises just above \$1.00. DARK-ECNs' market share of trades in stocks priced just above \$1.00 is almost double that of stocks priced just below \$1.00, which is robust to various measures of market share. The rise in DARK-ECN market share also corresponds to a discontinuous fall in the market share of trades of traditional exchanges.

The effects of the minimum pricing increment on intermarket competition are not limited to penny stocks. Minimum pricing increments also cause large buildups of liquidity demand for higher priced stocks constrained by tick size. When an exchange-listed stock has a large order depth and a narrow spread, a newly submitted order must be priced more aggressively to gain time priority over existing orders in the order book (Buti, Rindi, and Werner, 2011a). Otherwise, the new order is relegated to the back of the limit order queue.

For a random sample of 116 stocks stratified by market capitalization, we estimate a predicted bid-ask spread for each stock in the absence of a minimum pricing increment using factors shown in the market microstructure literature to affect spreads. The strength of the spread constraint in a stock is estimated by the observed bid-ask spread minus its predicted value and has a lower bound of zero. We find that the market share of dark pools increases with the strength of the spread constraint. Using stock depth to measure the size of limit order queues, we find evidence that on trading days when stocks are severely constrained by

[2] Dark pools operate much like traditional exchanges with the additional benefit of pre-trade opacity. Reg NMS is designed to protect top of the book quotes that are part of the National Best Bid and Offer (NBBO) from trade-throughs. An exchange that receives an order must either execute it at the NBBO or forward it to a market currently displaying the NBBO unless there is an exception to SEC Rule 611, the Order Protection Rule. However, Reg NMS does not protect time priority, so dark pools can enable queue jumping off exchanges.

[3] DARK-ECN and BLOCK venues constitute dark pools, and PING venues offer services that are analogous to those offered by dark pools (see SEC Concept Release No. 34-61358). We refer to retail market makers more generally as INTERNALIZE, as some broker-dealers that internalize orders also fall into this category.

[4] Trades on exchanges can occur on the half-penny through hidden limit orders pegged to the NBBO midpoint. In 2007 NYSE Arca was granted SEC permission to offer a midpoint order that executes in half-penny increments.

[5] Regression discontinuity analysis is used in Rauh (2006), Chava and Roberts (2008), Roberts and Sufi (2009), Iliev (2010), and Bakke and Whited (2012). Also see the survey on methodology in corporate finance by Roberts and Whited (2012).

the minimum pricing increment (exhibiting longer order queues), DARK-ECNs experience gains in market share.

One of the main disadvantages of trading in dark pools is the risk of nonexecution. Theoretical models argue for a trade-off between potential price improvement and the probability of nonexecution (see Zhu, 2014; Ye, 2011). In these models, traders face two types of transaction costs: price impact and nonexecution risk. Informed traders have an incentive to trade in a dark pool because orders submitted to dark pools typically have a lower price impact compared with orders submitted to an exchange. This is because dark pools match orders at or within the midpoint of the exchange bid and ask prices. But, no market maker exists to clear order imbalances, so buy and sell orders are often unbalanced, resulting in execution uncertainty. Spread constraints resulting from the minimum pricing increment encourage some traders, such as those with time-sensitive information, to bypass exchanges' long limit order queues that can delay execution by submitting their orders to dark venues. As traders shift their order flow to dark venues, the risk of order nonexecution falls. Thus, a positive liquidity externality emerges as traders are more likely to submit orders to dark pools when they perceive rising liquidity at this trading venue (Buti, Rindi, and Werner, 2011a). The positive feedback in trading is a potentially important driver of dark pool growth in recent years.

Our study is most closely related to theoretical papers examining the choice between exchange and dark pool trading venues. The Zhu (2014) model predicts that exchanges are more attractive to informed traders and dark pools are more attractive to uninformed traders. The rationale is that informed traders tend to cluster on one side of the market and, thus, face higher execution risk in dark pools because they have no market makers to absorb excess order flow. Liquidity traders are less likely to cluster on one side of the market and, hence, have a higher probability of execution in a dark pool.

Buti, Rindi, and Werner (2011a) build a theoretical model of a limit order market in which traders can choose to submit orders to the fully transparent limit order book or to a dark pool. For liquid stocks with high limit order book depth, they predict that tighter spreads in the dark pool foster price competition as traders can undercut the existing price in the limit order book, thus raising dark pool market share. When the limit order book has high depth and a narrow spread, new orders have to be more aggressively priced to gain priority over the existing orders in the book. Their model shows that dark pool market share is higher when tick size is larger. This is because, with larger tick sizes, market orders on exchanges are more expensive and traders have an incentive to trade within the spread in the dark pool.

Buti, Rindi, Wen, and Werner (2011) model the intermarket competition between a public limit order book and an internalization pool dark venue, which offers a smaller tick size. Market orders sent to the public limit order book are intercepted by the internalization pool in which better prices are available, resulting in a decline in liquidity demanded from the limit order book. Our study provides empirical support for many predictions of these theoretical models.

Concern is widespread that dark trading could be harming market quality. Studies investigating this issue examine the relation between dark trading, price discovery, and transaction costs. Theoretical studies model an informed trader's choice between trading in a dark pool and a traditional exchange, but they draw conflicting conclusions depending on the model parameters and type of private information assumed (see Hendershott and Mendelson, 2000; Buti Rindi, and Werner, 2011a; Ye, 2011; Zhu, 2014). Empirical studies investigate the impact of dark trading on market quality, and they often arrive at conflicting assessments, depending on data quality and research methods used (see O'Hara and Ye, 2011; Weaver, 2011; Buti, Rindi, and Werner, 2011b; Degryse, de Jong, and van Kervel, 2011; Comerton-Forde and Putnins, 2012; Foley, Malinova, and Park, 2012; Gresse, 2012; Nimalendran and Ray, 2014).

Equally important to evaluating the effects of dark market fragmentation is an understanding of why markets fragment. Markets fragment for a variety of reasons. Upstairs brokers can screen and facilitate trades for large institutional clients, who might not otherwise trade (see Seppi, 1990; Grossman, 1992). Barclay, Hendershott, and McCormick (2003) find that electronic communication networks (ECNs) offer the advantages of anonymity and speed of execution that attract informed traders away from Nasdaq market makers. We show that dark pools could increase the execution probability of limit orders placed relatively late. In general, limit order traders face a trade-off between possible improvement in execution price and the risk of nonexecution (see Foucault, 1999). With a buildup of depth on the exchange, the risk of nonexecution of new exchange orders increases. Dark pools offer selected traders the ability to bypass long queues of orders on a traditional exchange and more quickly execute their trades at or within the NBBO. This trading rule, which gives a competitive advantage to dark pools, is exogenously determined by the market regulator. Positive liquidity externalities resulting from this rule lead to increased market fragmentation, which is detrimental to market quality when one competitor is given a regulatory advantage over another.

Our findings also have implications for market transparency. The market share of trading in transparent markets has declined rapidly since the adoption of Reg NMS. Circumventing the time priority of displayed limit orders is likely to result in a withdrawal of liquidity providers from lit exchanges. Thus, to preserve market transparency, it is important to ensure that both transparent and opaque trading venues compete on an even playing field.

2. Institutional details

In this section, we compare the market features of traditional exchanges and dark venues and also describe our market venue classifications.

2.1. Exchanges versus dark venues

Dark venues offer several advantages over traditional exchanges: pre-trade transparency, preferential access, and subpenny price improvement.[6] Although off-exchange

[6] Reg ATS allows two main exemptions for alternative trading systems that execute less than 5% of the average daily volume in a

liquidity can come from either alternative trading systems or broker-dealer internalization, we discuss both sources of liquidity together because broker-dealers can offer many services analogous to those offered by ATSs.[7]

Our study primarily focuses on the third advantage, namely, the potential for subpenny price improvement. The Minimum Pricing Increment Rule (Rule 612) prohibits exchanges, alternative trading systems, and broker-dealers from displaying, ranking, or accepting a bid or offer, an order, or an indication of interest in an increment smaller than a penny for NMS stocks that are priced above $1.00. While all trading venues must abide by the display, rank, and acceptance conditions of Rule 612, broker-dealers that operate ATSs and those internalizing orders can offer price improvement in subpenny increments, resulting in trade executions within a penny. Currently, no requirement exists for trading centers to route orders to venues setting the NBBO quotations.[8] This allows dark venues to trade ahead of displayed limit orders with little or no price improvement.

Thus, competing venues offer a range of services that cater to different investor clienteles. For example, some dark venues choose to offer limited pre-trade transparency to a select group of investors through indications of interests (IOIs), which inform recipients of trading interest in the dark pool at the best quoted price or better.[9] Other venues offer payments to retail brokers for order flow deemed to be uninformed or they limit access to investors considered uninformed. Dark electronic communication networks, which operate much like exchanges, can compete on speed or subpenny executions when spreads on the exchanges are constrained at the penny increment. We focus on intermarket competition between trading venues when spreads are constrained.

While not the focus of this study, exchange venues can compete with other exchanges for order flow. A large number of studies investigate the competition for order flow between NYSE and Nasdaq (for example, see Christie and Huang, 1994; Huang and Stoll, 1996; Bessembinder and Kaufman, 1997). Similarly, Barclay, Hendershott, and McCormick (2003) and Huang (2002) investigate competition between Nasdaq market makers and electronic communication networks, which offer advantages of anonymity and execution speed.

2.2. Venue classifications

Nonexchange trading venues have a diverse range of operational structures (O'Hara and Ye, 2011). Mittal (2008) devises a five-way categorization of dark pools based largely on ownership structure: public crossing networks, internalization pools, ping destinations, exchange-based pools, and consortium-based pools. Zhu (2014) uses a three-way classification based on price discovery and order composition, namely, pools that match at the NBBO price midpoint, nondisplayed limit order books (DARK-ECNs), and electronic market makers. Rosenblatt Securities provides trading statistics for dark pools, although classified differently into bulge bracket, market maker, independent/agency, and consortium-sponsored pools.

In our data set, trades are classified into four main types: dark electronic communication networks (DARK-ECN), block crossing networks (BLOCK), ping destinations (PING), and retail market makers (INTERNALIZE), which are described below. Similar to Zhu (2014), these categories are based on operational structure, not ownership structure. Broker-to-broker negotiated trades and trades that cannot be categorized into any of the above four main types are classified as OTHER.[10] Thus, OTHER can include some trades that belong in the other four types.[11]

DARK-ECNs are operated by broker-dealers [Reg ATS Rule 301(b)(1)]. The operator could act either as an agent, by crossing institutional agency flow provided by their customers, or as a principal, by contributing liquidity through its market-making arm or proprietary trading desk. Additional liquidity is provided by external liquidity partners, who are typically high-frequency trading firms. DARK-ECN operators are also permitted to exclude sell side firms from accessing the pool. DARK-ECNs operate much like nondisplayed limit order books and accept both limit and market orders. Limit orders submitted to DARK-ECNs can execute ahead of displayed orders on transparent exchanges as long as the price is at or within the NBBO. Orders can also be submitted with a level of price improvement, which reduces the spread. We conjecture that traders use DARK-ECNs when spreads are constrained on major exchanges to enable them to jump the queue of existing displayed limit orders, while providing little or no price improvement.

BLOCK venues are the most traditional form of dark pool and are modeled widely in the finance literature (see Buti, Rindi, and Werner, 2011a; Ye, 2011; Zhu, 2014). Broker-dealers operating BLOCK venues act as an agent, with no proprietary order flow provided by the operator. Typically, these venues operate as continuous crossing networks matching buy and sell orders as they arrive at a price derived from the NBBO (usually at the midpoint) and charge a commission for this trading. The main advantage of BLOCK venues is the ability to execute large orders anonymously and with minimal price impact.

PING venues accept only immediate or cancel (IOC) orders from customers. IOC orders interact directly with the operator's proprietary order flow. The operator can

(*footnote continued*)
national market system stock during at least four of the six preceding calendar months. First, ATSs under the 5% threshold are exempt from providing their best-priced orders for inclusion in the consolidated quotation data for that security [Rule 301(b)(3)]. Second, an ATS that falls below the threshold can prohibit or limit any person from accessing the services offered by the ATS [Rule 301(b)(5)].

[7] See SEC Concept Release No. 34-61358.

[8] The SEC is debating a trade-at rule that would require incoming orders to be executed with significant price improvement or be routed to trading venues displaying the NBBO.

[9] The SEC has raised concerns that IOIs could be creating a two-tiered market if they are displayed only to selected market participants (see http://www.sec.gov/news/press/2009/2009-223-fs.htm).

[10] Bessembinder and Venkataraman (2004), Booth, Lin, Martikainen, and Tse (2002), Fong, Swan, and Madhavan (2001), Madhavan and Cheng (1997), and Smith, Turnbull, and White (2001) study broker-to-broker negotiated trades in upstairs markets.

[11] Trading platforms have many different features (Nimalendran and Ray, 2014), so classifying all trades is difficult.

accept incoming IOC orders, in which case the order trades against the operator's proprietary flow, or reject the order, in which case the order is canceled. While PING destinations can attract order flow by providing price improvement, the amount of price improvement is unknown to the trader at the time the order is submitted. In contrast to DARK-ECNs, in which the operator can act as either agent or principal, PING operators always trade as principals. Because PING operators actively choose which orders they trade against, we do not expect PING destinations to compete aggressively on the basis of queue jumping.

INTERNALIZE are retail market makers who internally match order flow coming from the customers of retail brokerage firms. Retail brokerages are often affiliated with a retail market maker who could pay these retail brokers for order flow.[12] Differences in market structures indicate that off-exchange venues compete among themselves and with exchanges in many ways.

2.3. Post-trade reporting

Off-exchange trades are reported first to a trade reporting facility (TRF), which then submits a report to the consolidated trade data feeds.[13] In 2011, only two TRFs—Financial Industry Regulatory Authority (FINRA)/Nasdaq and FINRA/NYSE—are active for most stocks. FINRA rules require a reporting member to submit information on trades including symbol, number of shares, trade price, and execution time.[14] For most transactions, trade reporting must be completed within 30 seconds of trade execution.[15] The trade report is then disseminated to the public via the consolidated tapes under the generic participant identifier D. Trades in NYSE- and AMEX-listed stocks are reported to the Consolidated Tape System (CTS), and trades in Nasdaq-listed stocks are reported to the Unlisted Trading Privileges (UTP) trade data feed.

3. Data and sample

We identify NYSE- and Nasdaq-listed stocks that cross the $1.00 threshold at least once over the period January 1, 2010 to March 31, 2011. To ensure variability in trade prices, we select an event window of 11 days: five days before the cross, the day of the cross, and five days after the cross. This event window allows us to differentiate between stocks that experience a permanent price decline or increase from stocks that fluctuate around one dollar.

Two trade reporting facilities are active during our sample period: FINRA/Nasdaq and FINRA/NYSE. Detailed trade data are obtained from the FINRA/Nasdaq TRF. For each transaction, there is information on the stock symbol, execution date, trade report time, trade price and volume, and a field that identifies the type of dark venue (i.e., DARK-ECN, BLOCK, PING, INTERNALIZE, or OTHER) on which the trade originated. Trade reports are time stamped to 10 milliseconds and represent the time at which the trade is reported to the TRF.

Dark trades reported to the FINRA/NYSE TRF are identified through the Securities Information Processor (SIP), which consolidates trade and quote data for public dissemination to trading participants. FINRA/NYSE TRF transactions are identified with participant identifier D and submarket code N, but without further identification of the origin of the dark trade (see Weaver, 2011). Trading venues reporting to the FINRA/NYSE TRF are likely to operate as DARK-ECNs, and we classify their trades accordingly.[16]

Trade data are processed to remove erroneous and irregular trades. Only regular trades executed between 9:30:30 and 16:00:00 are included. A 30-second delay from the 09:30 opening time is used to avoid contaminating our sample by the opening call auction. Regular trades are identified with condition codes @, @F, and F. This filter eliminates volume-weighted average price (VWAP) trades as these trades do not reflect how market participants respond to intraday market conditions. For reported trades, observations with account status codes B, C, D, E, I, K, N, and X are deleted, as these represent trade print backs, canceled, and broken trades. Daily time-weighted quoted spreads are calculated from SIP NBBO files, and market capitalization values are calculated from the Center for Research in Security Prices data base (CRSP).

Table 1 reports summary statistics for our sample of 1,060 events by 173 unique companies. *Mcap* is measured on day 0 of the event window and is expressed in millions of US dollars. All other variables are measured on a daily basis for trading days -5 to $+5$. The summary statistics show that we are dealing with a sample of small cap companies. The median stock has a *Mcap* of $26.01 million and trades 193,900 shares per day. However, large differences between mean and median values indicate that the data are highly skewed. The median *Qspread* is 1.6 pennies, indicating that many stocks in the sample could be constrained by the minimum pricing increment when the trade price is above a dollar.[17]

Table 1, Panel B, reports measures of trading activity for each of the trading venue types. Again, the data are skewed, indicated by differences between mean and median *Volume* and *Ntrades* values. Counter to our expectations, we find relatively large trade sizes for INTERNALIZE. This could be because retail orders, unlike institutional orders, are not sliced into smaller trade sizes by algorithms. Further, retail

[12] See SEC Concept Release No. 34-61358. Payment for retail order flow has been studied extensively in the literature (see Easley, Kiefer, and O'Hara, 1996; Bessembinder and Kaufman, 1997; Battalio, 1997).

[13] The Consolidated Tape Association (Nasdaq OMX) manages the collection, processing, and dissemination of trade and quote data for NYSE- and AMEX- (Nasdaq-) listed securities.

[14] See ⟨http://finra.complinet.com/en/display/display_main.html?rbid=2403&element_id=4392⟩. Reporting party and contra party market participant identifiers (MPIDs) are removed from our data set.

[15] See ⟨http://www.finra.complinet.com/en/display/display_viewall.html?rbid=2403&element_id=4439&record_id=5533⟩.

[16] See ⟨http://www.goldmansachs.com/media-relations/comments-and-responses/archive/market-structure-folder/alt-trading-sys.html⟩. Separating NYSE-TRF reported trades from DARK-ECN (not shown) yields similar conclusions.

[17] Furthermore, for trading days with an average price equal to or above $1.00, we find that 54.5% of days have a bid-ask spread of a penny occurring over a majority of the trading day. If we include the time when the bid-ask spread is at two pennies, the percentage of trading days increases to 74.7%.

Table 1
Summary statistics.
The table reports statistics for the 173 stocks in our sample. For the period January 1, 2010 to May 31, 2011, we select stocks that move from having a closing price below \$1.00 to above \$1.00 and vice versa. We analyze trade data for the 11-day event window from day −5 to +5 (without replacement), where day 0 is the first day the closing price of the stock is above or below \$1.00. Panel A and Panels B–D report statistics for stock and venue characteristics, respectively. Panel C (Panel D) reports venue characteristics for trading days when the average daily trade price is equal to or above \$1.00 (below \$1.00). *MCap* is the stock's market capitalization on day 0 in millions of dollars. If a stock crosses the \$1.00 threshold more than once, we average *MCap* across all events for the stock. *Price* is the average trade price in dollars. *Qspread* is the time-weighted average difference between the National Best Bid and Offer prices in dollars. *Volume* is the average daily volume in ten thousand share units. *Ntrades* is the average daily number of transactions. *TradeSize* is the average daily trade size. *Mktshare* is the number of trades for the indicated venue type divided by the total number of trades across all venue types. EXCH=exchange; DARK-ECN=dark electronic communication network; BLOCK=block crossing networks; PING=ping destinations; INTERNALIZE=retail market makers; OTHER=others.

Panel A: Stock characteristics

Variable	Number of observations	Mean	Standard deviation	Q1	Median	Q3
MCap (millions)	171	93.60	343.4	13.29	26.01	57.38
Price	173	1.195	1.292	0.988	1.016	1.065
Qspread	173	0.030	0.036	0.009	0.016	0.035
Volume	173	210.3	870.6	4.679	19.39	72.81
Ntrades	173	1,981	5,240	83.38	351.4	1,362
TradeSize	173	623.6	446.6	457.4	571.9	640.2

Trading venue	*Volume*		*Ntrades*		*TradeSize*		*Mktshare*	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Panel B: Venue characteristics (all stocks)								
EXCH	118.9	11.41	1,433	254.9	502.9	465.1	62.08	60.38
DARK-ECN	8.490	0.346	124.2	9.103	341.7	322.5	1.948	1.730
BLOCK	0.300	0.002	2.560	0.071	2,266	467.7	0.143	0.009
PING	17.72	1.534	156.4	23.40	736.5	717.1	9.459	9.472
INTERNALIZE	55.32	3.323	191.5	33.71	1,323	1,145	22.34	22.36
OTHER	9.561	0.725	73.18	10.63	1,117	715.0	4.032	3.918
Panel C: Venue characteristics (price ≥ \$1.00)								
EXCH	122.3	12.85	1,589	325.8	503.5	447.3	60.35	59.20
DARK-ECN	14.40	0.468	210.2	14.02	333.9	305.3	2.506	1.905
BLOCK	0.362	0.000	3.084	0.000	3,103	403.6	0.159	0.000
PING	20.71	2.152	201.6	31.33	717.0	719.0	9.762	9.586
INTERNALIZE	52.68	3.748	265.1	42.93	1,196	1,097	23.00	23.36
OTHER	11.08	1.102	98.20	12.38	1,562	672.3	4.230	4.026
Panel D: Venue characteristics (price < \$1.00)								
EXCH	123.6	8.931	1,374	183.0	515.9	475.4	64.19	62.20
DARK-ECN	3.378	0.182	48.78	5.11	360.9	342.6	1.375	1.221
BLOCK	0.247	0.000	2.159	0.000	1,766	429.2	0.137	0.000
PING	16.38	1.289	125.47	19.85	777.4	707.6	9.105	9.267
INTERNALIZE	61.75	2.602	129.13	22.17	1,490	1,235	21.34	21.95
OTHER	8.453	0.556	52.23	8.71	1,081	650.1	3.851	3.597

trading involves a large amount of activity by day traders and high net worth investors, both of whom can generate large orders. Less sophisticated retail traders are likely to be investing in stocks via mutual funds and pension fund accounts so that these trades would reflect typical institutional trading behavior.[18]

For each stock, we define *Mktshare* as the share volume occurring in the indicated venue type divided by the total share volume across all venue types. Of the dark venues, INTERNALIZE reports the highest *Mktshare*, which is consistent with the notion that retail investors are drawn to highly volatile, low-priced stocks. BLOCK venues are inactive in dollar stocks, reflecting the low execution probabilities in these venues.

Table 1, Panels C and D, summarizes the venue characteristics for trading days when the average trade price is above and below \$1.00, respectively. When stocks are priced above \$1.00, we observe higher *Mktshare* for the dark trading venues and lower exchange (EXCH) *Mktshare* compared with when the stocks are priced below \$1.00. DARK-ECN shows the largest percentage change in *Mktshare*, indicating that this trading venue could offer some trading advantages at stock prices above \$1.00. In the next section, we use changes in *Mktshare* to proxy for changes in trading venue competitiveness.

4. Empirical results

In this section, we first present figures showing the relationship between stock prices, trading volume and market share around the \$1.00 price threshold. Then, we formally test the relationship between stock price and

[18] Some large broker-to-broker negotiated trades or BLOCK trades also could be misclassified as INTERNALIZE.

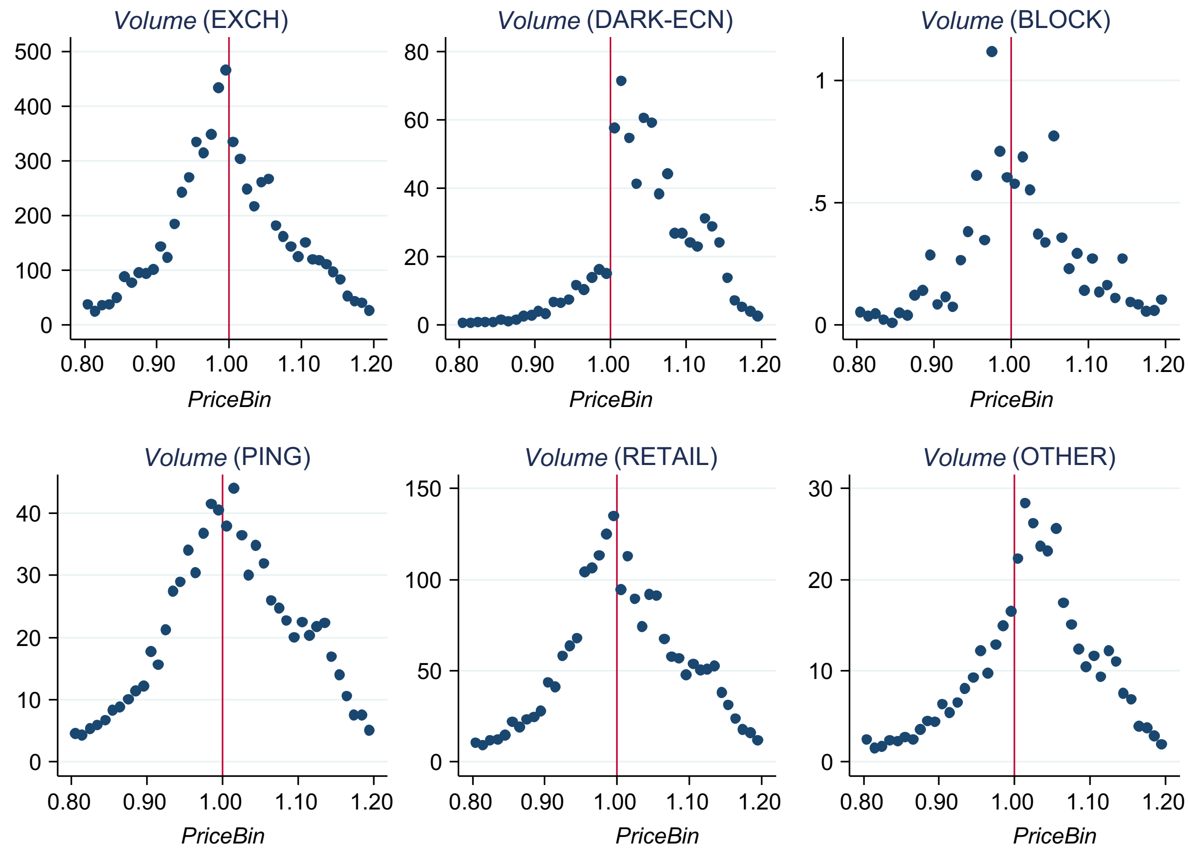

Fig. 1. Regression discontinuity plots: *Volume* against *PriceBin.* This figure plots *Volume* against *PriceBin* for each trading venue type: exchange (EXCH), dark electronic communication networks (DARK-ECN), block crossing networks (BLOCK), ping destinations (PING), retail market makers (INTERNALIZE), and others (OTHER). For each penny price bin from \$0.80 to \$1.20, we find the total number of shares traded (*Volume* in millions). The vertical line in each plot indicates the \$1.00 price level.

market share using discontinuity design and first difference regressions.

4.1. Trading around \$1.00 threshold

Fig. 1 plots trading volume against price. All trades are classified such that $\$0.80 \leq \text{price} < \$0.81$ are in the \$0.80 price bin (*PriceBin*) and so forth. We show plots for each trading venue type for trades between \$0.80 and \$1.20. The vertical line in each plot indicates the \$1.00 threshold at which the minimum pricing increment changes. In the absence of a discontinuity, we expect the distributions to appear as a smooth triangle, which peaks at \$1.00. We expect lower trade activity at the ends of the price distribution because, while all stocks cross the \$1.00 threshold, not all stocks fall in price to \$0.80 or rise to \$1.20 over the 11-day sample period.

Our hypothesis predicts a sharp rise in volume for dark venues and a sharp decline in volume for traditional exchanges when the stock price crosses above \$1.00. Consistent with our predictions, we observe a decrease in EXCH volume and a positive jump in DARK-ECN and OTHER volume. We also observe a slight decrease in INTERNALIZE volume. In contrast, there are no visible discontinuities in plots for BLOCK and PING, indicating that trading in these venues does not exhibit sensitivity to the spread constraint.

To test for changes in relative competitiveness of trading venues, we investigate changes in *Mktshare* around the \$1.00 threshold. Fig. 2 plots *Mktshare* against *PriceBin*. For each stock, we calculate *Mktshare* for every trading venue type at each *PriceBin* and find the cross-sectional mean. *Mktshare* is defined for a given stock as the share volume in a trading venue type divided by the total share volume in the *PriceBin* across all venues.[19] We present plots for each trading venue type for trades at prices between \$0.80 and \$1.20.

We predict a decline in *Mktshare* for EXCH and a rise in *Mktshare* for dark venues when the trade price rises above \$1.00. For EXCH (DARK-ECN) we observe a negative (positive) jump in *Mktshare* for trades just above \$1.00 compared with trades just below \$1.00. The plot shows a jump in DARK-ECN *Mktshare* from around 1.5% to over 2.5% when the price crosses above \$1.00. We do not observe a discontinuity effect for BLOCK, PING, and INTERNALIZE, indicating that these venues are insensitive to this price

[19] To reduce outlier effects, we first remove the largest 1% of trades and require each trading bucket to have at least 20 trades.

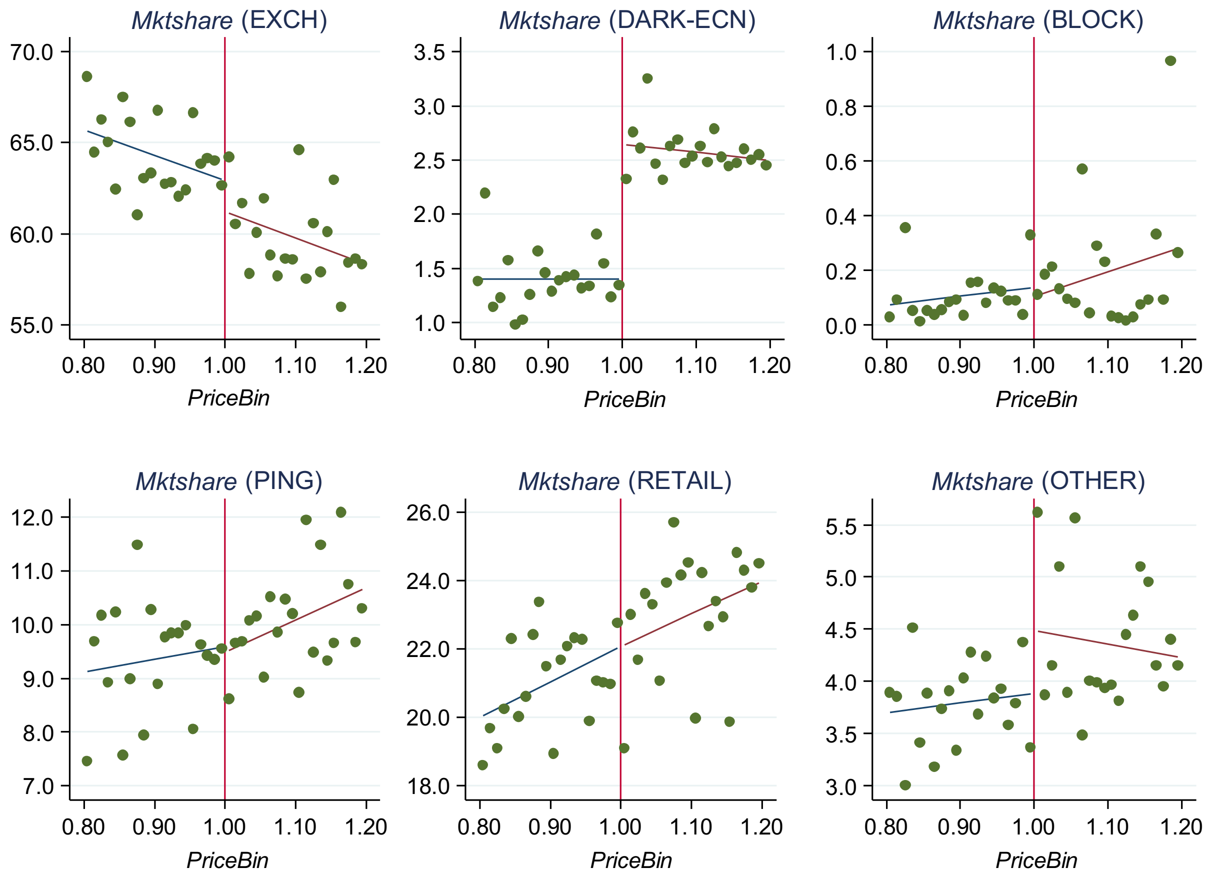

Fig. 2. Regression discontinuity plots: *Mktshare* against *PriceBin*. This figure plots *Mktshare* against *PriceBin* for each trading venue type: exchange (EXCH), dark electronic communication networks (DARK-ECN), block crossing networks (BLOCK), ping destinations (PING), retail market makers (INTERNALIZE), and others (OTHER). For each penny price bin from \$0.80 to \$1.20, we find the *Mktshare* for each stock, with *Mktshare* calculated as the number of shares traded for the trading venue type as a percentage of the total number of shares traded in the price bin. These plots show the mean *Mktshare* for each *PriceBin* across all stocks. The vertical line in each plot indicates the \$1.00 price level.

threshold. There is a small discontinuity for OTHER, although the jump is less pronounced than for DARK-ECN, indicating that some trades in the OTHER category could be misclassified in terms of their trading venue. Similar plots are obtained if *Mktshare* is calculated based on number of trades, instead of share volume.

Next, we perform a discontinuity design regression on a per stock basis to test the significance of jumps in *Mktshare* more formally. A major advantage of discontinuity design regressions (DDRs) is that only mild assumptions are required for causal inferences compared with other nonexperimental approaches (Lee and Lemieux, 2010). For a DDR analysis to be as credible as a randomized experiment, we need to assume only that the density function of the treatment variable is continuous across a known threshold point (Lee, 2008). Thus, causal inferences drawn from a DDR can be more credible than inferences drawn from other natural experiment designs, such as difference-in-differences or instrumental variable techniques, which require stronger assumptions to be valid. The validity of a DDR experiment relies on the randomness of a stock price crossing above or below the \$1.00 threshold. We assume that traders cannot precisely manipulate the price around \$1.00, and, thus, stock prices cross the \$1.00 threshold randomly.[20]

We create price bins by rounding each trading price downward to the nearest whole cent, so that all the trades with a price from \$0.80 to less than \$0.81 are in the \$0.80 price bin. We typically use a subset of these bins such that $\$1.00-h \le PriceBin \le \$1.00+h$. For $h=\$0.20$ there are 41 bins ranging from \$0.80 to \$1.20. We estimate the following pooled regression around the \$1.00 threshold and bandwidth, h:

$$Mktshare_{i,k} = \beta_0 + \delta_0\, DollarPrice_{i,k} + \beta_1 (PriceBin_{i,k} - \$1.00) + \beta_2\, DollarPrice_{i,k}(PriceBin_{i,k} - \$1.00) + \epsilon_{i,k}, \tag{1}$$

where, for stock i and price bin k, $Mktshare_{i,k}$ is the share volume in the indicated trading venue divided by the total share volume across all trading venues, $DollarPrice_{i,k}$ is an

[20] We formally test this underlying assumption of our DDR framework. Specifically, we test whether the number of trades across all trading venues changes around the \$1.00 threshold and find a statistically insignificant change.

Table 2
Regression discontinuity tests: market share.

The table reports regression discontinuity tests for market share. For the period January 1, 2010 to May 31, 2011, we select stocks that move from having a closing price below \$1.00 to above \$1.00 and vice versa. We analyze trade data for the 11-day event window from day -5 to $+5$, where day 0 is the first day the closing price of the stock is either above or below \$1.00. Trades are grouped into penny price bins and classified into six types: exchange (EXCH), dark electronic communication networks (DARK-ECN), block crossing networks (BLOCK), ping destinations (PING), retail market makers (INTERNALIZE), and those not previously classified, (OTHER). The results are based on the following linear regression around the \$1.00 threshold:

$$Mktshare_{i,k} = \beta_0 + \delta_0\, DollarPrice_{i,k} + \beta_1\,(PriceBin_{i,k} - \$1.00) + \beta_2\, DollarPrice_{i,k}(PriceBin_{i,k} - \$1.00) + \epsilon_{i,k},$$

where $Mktshare_{i,k}$ is the number of shares traded for the indicated trading venue type as a percentage of the total number of shares traded, for stock i and price bin k. k represents the lower bound of the price bin. For example, trades priced at \$1.00 to \$1.0099 are contained in $PriceBin_{i,1.00}$. $DollarPrice_{i,k}$ is an indicator variable equal to one if k is equal to or greater than \$1.00. Panels A and B report results for trading between \$0.80 and \$1.20 and \$0.90 and \$1.10, respectively. All regressions control for stock fixed effects. We report heteroskedasticity-robust standard errors clustered by stock in parentheses. ***, **, and * indicate a significance level of 1%, 5%, and 10%, respectively.

Variable	EXCH	DARK-ECN	BLOCK	PING	INTERNALIZE	OTHER
Panel A: \$0.80 ≤ PriceBin ≤ \$1.20						
DollarPrice	−2.244***	1.389***	−0.010	0.102	0.396	0.367
	(0.77)	(0.24)	(0.05)	(0.40)	(0.61)	(0.28)
PriceBin − \$1.00	−12.697*	−0.092	0.294	2.897	9.445*	0.153
	(7.25)	(1.34)	(0.48)	(3.63)	(5.68)	(2.23)
DollarPrice × (*PriceBin* − \$1.00)	2.785	−1.091	−0.186	1.263	−2.520	−0.251
	(10.04)	(1.83)	(0.58)	(4.53)	(7.88)	(2.97)
Constant	62.303***	0.533***	0.106**	11.691***	20.055***	5.312***
	(0.69)	(0.15)	(0.04)	(0.33)	(0.54)	(0.24)
Number of observations	4,861	4,861	4,861	4,861	4,861	4,861
Adjusted R-squared	0.232	0.259	0.014	0.125	0.217	0.087
Panel B: \$0.90 ≤ PriceBin ≤ \$1.10						
DollarPrice	−2.412**	1.167***	−0.007	0.040	0.488	0.724**
	(0.93)	(0.26)	(0.07)	(0.45)	(0.72)	(0.31)
PriceBin − \$1.00	−14.500	1.728	1.071	2.891	11.805	−2.995
	(9.95)	(1.97)	(0.65)	(5.93)	(9.09)	(4.00)
DollarPrice × (*PriceBin* − \$1.00)	8.899	−0.872	−1.773*	0.601	−4.138	−2.717
	(14.13)	(3.32)	(1.06)	(7.91)	(12.00)	(5.75)
Constant	60.090***	0.967***	0.214***	12.645***	19.975***	6.110***
	(0.63)	(0.15)	(0.04)	(0.36)	(0.54)	(0.23)
Number of observations	3,037	3,037	3,037	3,037	3,037	3,037
Adjusted R-squared	0.295	0.272	0.023	0.154	0.275	0.125

indicator that takes the value of one if the price bin is equal to or greater than \$1.00 and a value of zero if price bin is less than \$1.00, and ε is a random error term. δ_0 is the magnitude of the discontinuity at the \$1.00 threshold. Inferences are based on heteroskedasticity-consistent standard errors, clustered by individual stock.

Table 2, Panel A, reports regression estimates where h equals \$0.20. We find a statistically and economically significant 1.4% jump in DARK-ECN market share at the \$1.00 threshold. EXCH *Mktshare* declines by 2.2% when the stock price crosses above \$1.00.

The next step in regression discontinuity analysis is to choose the optimal bandwidth. The choice of bandwidth size is a trade-off between precision and bias. A larger bandwidth yields more precise coefficient estimates because more observations are used. However, estimates using larger bandwidths are biased if the model specification is nonlinear. Smaller bandwidths are more likely to produce unbiased estimates as a linear specification provides a close approximation even if the underlying model specification is nonlinear.

Table 2, Panel B, uses a bandwidth of \$0.10. Overall, the results are qualitatively similar to those in Panel A, although the precision of the estimate declines as bandwidth size declines. We find that EXCH *Mktshare* declines and DARK-ECN *Mktshare* increases as soon as the stock price crosses above \$1.00. Our results are robust to the addition of higher-order terms in the regression model and to subsamples based on listing market.

Taken together, these results show that circumventing the time priority of displayed limit orders is one way DARK-ECNs compete for order flow. Broker-dealers operating DARK-ECNs can act as both agents and principals. When the broker-dealer is acting solely as an agent, DARK-ECNs function much like hidden limit order books offering faster executions of orders.

We claim that any differences in market share on either side of the cutoff should be due only to the effects of spread constraints and the ability to trade within the NBBO. This claim relies on several identification assumptions. First, as we approach \$1.00 from either side, any differences in stock characteristics are assumed to be random, which is supported by the fact that a stock trading just above \$1.00 has similar risk characteristics as the same stock trading just below \$1.00. Second, we assume that there is no significant change in trader types around \$1.00.

The SEC defines penny stocks as speculative securities of very small companies priced below \$5, and thus,

Table 3
Regression discontinuity tests with additional controls: market share.

Table 3 reports regression discontinuity tests for market share, after controlling for *Ntrades* and *Tradesize*. For the period January 1, 2010 to May 31, 2011, we select stocks that move from having a closing price below $1.00 to above $1.00 and vice versa. We analyze trade data for the 11-day event window from day −5 to +5, where day 0 is the first day the closing price of the stock is above or below $1.00. Trades are grouped into penny price bins and classified into six types: exchange (EXCH), dark electronic communication networks (DARK-ECN), block crossing networks (BLOCK), ping destinations (PING), retail market makers (INTERNALIZE), and those not previously classified, (OTHER). We estimate the linear regression.

$$Mktshare_{i,k} = \beta_0 + \delta_0 DollarPrice_{i,k} + \beta_1(PriceBin_{i,k} - \$1.00) + \beta_2\, DollarPrice_{i,k}(PriceBin_{i,k} - \$1.00) + \beta_3 \,\mathrm{Ln}(Ntrades_{i,k}) + \beta_4 \,\mathrm{Ln}(TradeSize_{i,k}) + \epsilon_{i,k},$$

where $Mktshare_{i,k}$ is the number of shares traded for the indicated trading venue type as a percentage of the total number of shares traded, for stock *i* and price bin *k*. *k* represents the lower bound of the price bin. For example, trades priced at $1.00 to $1.0099 are contained in $PriceBin_{i,1.00}$. $DollarPrice_{i,k}$ is an indicator variable equal to one if *k* is equal to or greater than $1.00. $Ntrades_{ik}$ and $TradeSize_{ik}$ are the total number of trades and average transaction size across all trading venues, respectively. Panels A and B report results for trading between $0.80 and $1.20 and $0.90 and $1.10, respectively. All regressions control for stock fixed effects. We report heteroskedasticity-robust standard errors clustered by stock in parentheses. ***, **, and * indicate a significance level of 1%, 5%, and 10%, respectively.

Variable	EXCH	DARK-ECN	BLOCK	PING	INTERNALIZE	OTHER
Panel A: $0.80 ≤ PriceBin ≤ $1.20						
DollarPrice	−1.971**	1.492***	−0.017	0.108	0.057	0.331
	(0.78)	(0.24)	(0.05)	(0.39)	(0.62)	(0.28)
PriceBin − $1.00	−21.035***	−1.726	0.002	7.769**	15.479***	−0.488
	(7.47)	(1.35)	(0.45)	(3.58)	(5.71)	(2.37)
DollarPrice × (*PriceBin* − $1.00)	17.419	1.543	0.405	−8.067	−12.444	1.144
	(10.93)	(1.84)	(0.57)	(5.01)	(8.21)	(3.21)
Log(*Ntrades*)	1.215***	0.211***	0.052***	−0.801***	−0.802***	0.125
	(0.37)	(0.07)	(0.02)	(0.19)	(0.26)	(0.10)
Log(*TradeSize*)	−2.466***	−0.870***	0.042	0.151	2.887***	0.256
	(0.93)	(0.21)	(0.04)	(0.53)	(0.71)	(0.31)
Constant	69.494***	4.550***	−0.507	16.186***	7.417	2.859
	(6.92)	(1.51)	(0.32)	(3.84)	(5.20)	(2.15)
Number of observations	4,861	4,861	4,861	4,861	4,861	4,861
Adjusted *R*-squared	0.241	0.272	0.016	0.132	0.228	0.087
Panel B: $0.90 ≤ PriceBin ≤ $1.10						
DollarPrice	−2.331**	1.217***	−0.016	0.123	0.373	0.633**
	(0.95)	(0.27)	(0.06)	(0.44)	(0.75)	(0.32)
PriceBin − $1.00	−24.001**	0.591	0.813	8.785	18.642**	−4.830
	(9.79)	(1.90)	(0.64)	(6.04)	(8.68)	(3.97)
DollarPrice × (*PriceBin* − $1.00)	27.044*	1.051	−1.212	−11.480	−16.849	1.447
	(14.67)	(3.23)	(1.02)	(8.34)	(11.94)	(5.87)
Log(*Ntrades*)	1.966***	0.236**	0.053**	−1.218***	−1.416***	0.379
	(0.48)	(0.11)	(0.02)	(0.22)	(0.38)	(0.11)
Log(*TradeSize*)	−1.514	−0.496	0.045	−0.097	1.526**	0.536
	(1.01)	(0.31)	(0.04)	(0.67)	(0.76)	(0.45)
Constant	56.470***	2.512	−0.427	21.406***	19.836***	0.202
	(7.52)	(2.33)	(0.30)	(4.54)	(5.96)	(3.20)
Number of observations	3,037	3,037	3,037	3,037	3,037	3,037
Adjusted *R*-squared	0.307	0.279	0.024	0.167	0.286	0.128

changes in institutional trading interest should occur at $5.00, not $1.00.[21] Some mutual funds are restricted from investing in penny stocks. Third, it is costly to short sell stocks priced under $5.00 per share. For stocks priced under $5.00, FINRA Rule 4210 requires a maintenance margin, which is the greater of $2.50 per share or 100% of the stock's current market value. This means that the level of short selling or institutional interest is unlikely to change around the dollar threshold.[22] As further robustness, we include controls for total number of trades and trade size. The coefficient estimates reported in Table 3 are largely consistent in magnitudes and significance with those in Table 2. Again, we find, after controlling for trading characteristics, a discrete jump in DARK-ECN *Mktshare* as stock price crosses above $1.00.[23]

[21] See ⟨http://www.sec.gov/answers/penny.htm⟩.

[22] In further tests, we compare institutional ownership for quarters with a closing price between $0.80 and $1.00 and quarters with a closing price between $1.00 and $1.20, using Form 13F filings. We find no significant difference in institutional ownership levels (*p*-value = 0.232), nor in the percentage of stocks with no institutional ownership (*p*-

(*footnote continued*)
value = 0.768). Comparing changes in quarterly institutional ownership for stocks that rise in price from below $1.00 to above $1.00 (risers) and stocks that fall from above $1.00 to below $1.00 (decliners), we again find no significant difference in the percentage change in institutional ownership for risers and decliners (*p*-value = 0.526).

[23] The Internet Appendix shows a rise in dark trading after a reverse split from a pre-split stock price below a $1.00.

Table 4
First-differenced estimation: market share.

The table reports the results from first-differenced regressions for changes in market share. For the period January 1, 2010 to May 31, 2011, we select stocks that move from having a closing price below \$1.00 to above \$1.00 and vice versa. We analyze trade data for the 11-day event window from day −5 to +5, where day 0 is the first day the closing price of the stock is above or below \$1.00. Trades are grouped into penny price bins and classified into six types: exchange (EXCH), dark electronic communication networks (DARK-ECN), block crossing networks (BLOCK), ping destinations (PING), retail market makers (INTERNALIZE), and those not previously classified, (OTHER). The results are based on the first-difference regression equation.

$$\Delta Mktshare_i = \delta_0 + \beta_1 \, \Delta \, \mathrm{Ln}(Ntrades_i) + \beta_2 \, \Delta \, \mathrm{Ln}(Tradesize_i) + \beta_3 \, \Delta Volatility_i + \epsilon_i,$$

where $Mktshare_i$ is the daily number of shares traded in the trading venue type as a percentage of the total number of shares traded for event *i*, $Ntrades_i$ is the daily number of trades, $Tradesize_i$ is the average daily trade size, and $Volatility_i$ is the difference between the log of the intraday high price and the log of the intraday low price. All variables are first measured on a daily basis then averaged across the five trading days in the pre-event window and the five trading days in the post-event window (including the cross day). For each variable, we subtract the mean value for the trading days below \$1.00 from the mean of the trading days above \$1.00. Panels A–C present results for all events, single events and constrained single events, respectively. Single events contains only events in which the stock is consecutively above (below) \$1.00 from day −5 to −1 and then below (above) \$1.00 from day 0 to +4. Constrained single events is a subset of single events and contains events in which the time-weighted quoted spread is below \$0.011 on days when the time-weighted average bid-ask midpoint (*Midpoint*) is above \$1.00 and below \$0.01 on days when *Midpoint* is below \$1.00. We report heteroskedasticity-robust standard errors clustered by stock in parentheses. ***, **, and * indicate a significance level of 1%, 5%, and 10%, respectively.

Variable	EXCH	DARK-ECN	BLOCK	PING	INTERNALIZE	OTHER
Panel A: All events						
δ_0	−1.781***	0.433***	0.027	0.032	0.411	0.217*
	(0.30)	(0.08)	(0.03)	(0.18)	(0.36)	(0.12)
$\Delta Ntrades$	3.345***	0.075	0.032	−0.331	−2.556***	−0.132
	(0.59)	(0.08)	(0.03)	(0.20)	(0.52)	(0.16)
$\Delta TradeSize$	−0.716	−0.444	−0.017	0.007	−0.039	0.156
	(1.69)	(0.29)	(0.04)	(0.43)	(1.00)	(0.78)
$\Delta Volatility$	−7.412**	−0.978	−0.605	−4.394*	1.033	1.721
	(3.34)	(0.77)	(0.51)	(2.35)	(3.50)	(1.76)
Number of observations	1,060	1,060	1,060	1,060	1,060	1,060
Adjusted *R*-squared	0.051	0.009	0.001	0.010	0.034	−0.001
Panel B: Single events						
δ_0	−4.896***	1.104***	0.110	0.890**	2.735***	1.090***
	(0.68)	(0.24)	(0.10)	(0.38)	(0.79)	(0.30)
$\Delta Ntrades$	3.485***	0.029	0.037	−0.617	−2.879***	−0.453*
	(0.71)	(0.16)	(0.06)	(0.37)	(0.72)	(0.27)
$\Delta TradeSize$	−1.643	−1.131	−0.058	0.895	−0.383	1.690**
	(2.19)	(0.81)	(0.12)	(0.95)	(1.80)	(0.69)
$\Delta Volatility$	−8.274***	−0.406	−0.568	−2.444*	5.270	2.702
	(2.91)	(1.08)	(0.69)	(1.45)	(4.62)	(1.77)
Number of observations	279	279	279	279	279	279
Adjusted *R*-squared	0.094	0.021	−0.008	0.018	0.048	0.025
Panel C: Constrained single events						
δ_0	−5.417***	2.146***	0.021	1.104*	0.910	1.310***
	(0.90)	(0.60)	(0.21)	(0.55)	(1.61)	(0.35)
$\Delta Ntrades$	4.188***	0.793*	0.035	−1.144	−3.722**	−0.098
	(1.39)	(0.40)	(0.10)	(0.70)	(1.62)	(0.29)
$\Delta TradeSize$	−5.875**	−3.506	0.161	0.607	12.420**	−1.518
	(2.74)	(2.52)	(0.46)	(0.96)	(5.03)	(0.97)
$\Delta Volatility$	−12.252**	−4.152	−0.292	4.733	2.753	4.001***
	(6.04)	(2.80)	(0.67)	(2.89)	(8.38)	(1.25)
Number of observations	82	82	82	82	82	82
Adjusted *R*-squared	0.172	0.123	−0.037	0.003	0.140	0.021

4.2. First-differenced analysis

We repeat the prior experiment using a first-differenced approach based on daily data. In our experiment, first-differencing has some advantages over regression discontinuity analysis. Matching quotes and trades using intraday data introduces some timing issues, which can be overcome by using daily values. Estimating the average daily bid-ask spread allows us to compare stocks that on average are more spread constrained with stocks that are less spread constrained. Using daily data, we can also identify stocks that show a permanent change in price (i.e., remain above or below \$1.00 for an extended period of time).

For the daily analysis, we estimate the first-difference regression equation

$$\Delta Mktshare_i = \delta_0 + \beta_1 \, \Delta \, \log(Ntrades_i) + \beta_2 \Delta + \beta_3 \, \Delta Volatility_i + \epsilon_i, \tag{2}$$

where $Mktshare_i$ is the daily share volume for each trading venue type divided by the total share volume for pricing event *i*, $Ntrades_i$ is the daily number of trades, $Tradesize_i$ is the average daily trade size, and $Volatility_i$ is the difference between the log of the intraday high ask price and the log of the intraday low bid price. All variables are first measured on a daily basis. Next, we average the daily variables across the five trading days -5 to -1 in the pre-event window and the five trading days 0 to $+4$ in the post-event window.

Our analysis consists of two types of pricing events: events with prices that cross above $1.00 and events with prices that cross below $1.00. To account for the two scenarios, we subtract the mean value for the trading days when the closing price is below $1.00 from the mean value for the trading days when it is above $1.00, which ensures consistency in our interpretation of δ_0. This means that we subtract the post-event mean from the pre-event mean for events in which the stock price crosses below the $1.00 threshold.

Table 4, Panel A, shows the results for the full sample of events. Consistent with our main analysis, we find that δ_0 is negative and significant for EXCH, indicating that traditional exchanges lose market share on days when the stock trades above $1.00. In contrast, δ_0 is positive and significant for DARK-ECN and OTHER, meaning that these trading venue types are more competitive when the price is above $1.00.

The full sample results presented in Table 4, Panel A, use all event windows formed over the sample period. However, there could be overlapping events if stocks fluctuate around $1.00 over short time periods. In addition, we expect the strongest results when stock prices are constrained. We form subsamples for single events and constrained single events. The single event subsample contains events in which a stock crosses the $1.00 threshold once within the event window. The constrained single events subsample also requires that *Qspread* is below 1.1 pennies on trading days when the average daily stock price is above $1.00.[24]

Panels B and C present the results for the single event subsample and constrained single event subsample, respectively. Supporting our predictions, we find that δ_0 is consistently positive and significant for DARK-ECN and OTHER and negative and significant for EXCH across all panels. Importantly, the economic significance of our findings is largest for our subsample of constrained single events. This finding supports our prediction that DARK-ECNs are most beneficial to traders when exchange trading is constrained by minimum pricing increments as these venues can offer traders narrower tick sizes and higher time priorities.

5. Additional robustness and alternative explanations

We detail additional robustness tests and alternative explanations for our findings. For brevity, we report the tabulated results in the Internet Appendix.

5.1. Placebo tests for alternative price thresholds

In US markets, there is only one regulatory mandated change to the minimum pricing increment in our sample period, which occurs at $1.00. Thus, we should not observe a sharp decrease in EXCH trading corresponding to a sharp increase in DARK-ECN trading for other price cutoff points. We test for trading activity discontinuities around several alternative price thresholds, namely, at $2.00, $5.00, and $50.00.[25] We fail to observe a similar drop in EXCH trading at these alternative price cutoffs, further supporting our hypothesis.

5.2. Constrained and unconstrained stocks

If the spread constraint is an important factor driving trading venue competition, we should see the strongest results in stocks that are tightly constrained by the minimum pricing increment. We divide our sample into constrained and unconstrained stocks. A constrained stock has a *Qspread* below 1.1 pennies on days with an average trade price above $1.00 and experiences a reduction in the spread when the average trade price falls below $1.00. This means that the spreads of these stocks are tightly constrained when the price is above $1.00 and becomes unconstrained when the price falls below $1.00. The minimum pricing increment should have no impact on trading in unconstrained stocks, which we define as a stock that has a *Qspread* above 1.1 pennies, regardless of price. We expect the changes in *Mktshare* to hold only for the constrained stock sample, and the evidence supports our hypothesis. Specifically, we find for the constrained stocks that EXCH *Mktshare* significantly falls and DARK-ECN *Mktshare* significantly rises when the price crosses above the dollar threshold. For the unconstrained stock sample, these *Mktshare* changes are statistically insignificant.

5.3. Alternative measures of market share

In our main results, we calculate *Mktshare* for every stock and price bin combination and test for regression discontinuities using pooled ordinary least squares (OLS) regressions with stock fixed effects. Our summary statistics in Table 1 show that the data are highly skewed. As a robustness test, we construct volume-weighted measures of market share. For each *PriceBin*, we sum the share volume for each trading venue type and divide by the total share volume within the price bin. The results show a significant decrease (increase) in EXCH (DARK-ECN) market share when the stock price crosses above the $1.00 threshold.

Our results so far are based on market shares expressed in share volumes to capture trading venue competition. As an alternative measure of trading venue competition, we calculate market shares based on the number of trades, which captures a trader's decision on whether to trade in a particular venue, in contrast to trading volume, which captures a trader's decision on how much to trade. Again, we obtain qualitatively similar results to our main findings.

5.4. Logit regressions

An alternative methodology for testing trading venue competitiveness is to estimate the likelihood of a trade

[24] We use 1.1 pennies, instead of one penny, to allow for minimal variations in *Qspread* throughout the trading day. Otherwise, a single trade that walked the book could eliminate that stock from our sample.

[25] For this analysis, we use Trade and Quote (TAQ) data, which contains a generic identifier for TRF reported trades. TRF represents aggregate trades of DARK-ECN, BLOCK, PING, INTERNALIZE, and OTHER.

taking place in a particular trading venue type. We estimate a multinomial logit model in which the dependent variable, *Venue*, takes a value from zero to five, depending on the venue type in which the trade is executed, where zero to five represents EXCH, DARK-ECN, BLOCK, PING, INTERNALIZE, and OTHER, respectively. The independent variable is *DollarPrice*, an indicator that takes a value of one if the trade price is at or above $1.00, and we also control for *Tradesize* and *Price*. Our results are consistent with a migration of order flow away from EXCH when trading is constrained by minimum pricing increments to take advantage of queue jumping on DARK-ECNs. Furthermore, splitting trades into those that are constrained (i.e., prevailing bid-ask spread is at the minimum pricing increment) and unconstrained, we find that the results are driven by our sample of constrained trades.

5.5. Changes in a stock's risk (e.g., probability of delisting)

Stocks that fall below the $1.00 price level and remain at this level for a period of time risk being delisted from the exchange.[26] In our regression discontinuity design, we assume that a stock trading at $1.01 has essentially the same risk of delisting as a stock trading at $0.99. To further investigate the validity of this assumption, we split our events into stocks that fluctuate around $1.00 and stocks that consistently remain below (above) $1.00 for five consecutive days and then stay above (below) $1.00 for the next five days. We label these stocks permanent risers and decliners. While permanent risers (decliners) could experience a decrease (increase) in the probability of delisting, stocks persistently fluctuating around $1.00 are likely to experience a negligible change in their probability of delisting. The results show that a rise in delisting risk is unlikely to be driving our results. Specifically, we find that, across the three subsamples, *DollarPrice* is positive and significant for DARK-ECN and negative and significant for EXCH, which is consistent with the results from our main tests.

As an additional test, we identify NYSE- and Nasdaq-listed stocks having a minimum closing daily price between $0.80 and $1.00 (62 stocks) and $1.00 and $1.20 (64 stocks) during 2010. We then compare the number of exchange delistings that occur over the subsequent two-year period for the paired samples. Using a chi-square test, we find no significant difference in the number of delistings between the two samples (p-value$=0.203$). This is further evidence that the risk of stock delisting is not a key factor driving our results.

5.6. Maker-taker fee structure

Our results are unlikely to be driven by changes in exchange maker-taker fee structures around the $1.00 threshold. Across the main exchanges, the rebate for posting liquidity falls when a stock price drops below $1.00, discouraging liquidity providers from providing liquidity on the exchange, which should reduce exchange order flow. At prices immediately below the $1.00 threshold, there is also an increase in taker fees on the NYSE and Nasdaq for absorbing liquidity. For example, the taker fee on the NYSE is $0.0025 per share for stocks priced at or above $1.00 and rises to 0.3% for stocks priced below $1.00. Thus, when a stock price falls below $1.00, this fee structure should lead order flow to migrate away from more expensive exchange venues. Our empirical evidence shows just the opposite effect. We find order flow migrates to the exchange when the stock is priced below $1.00. In contrast, dark pools vary by venue and by investors in their pricing policies, though some venues make no change in fees around the $1.00 threshold (Harris, 2013).[27]

6. Further tests of venue competitiveness

The prior evidence indicates that dark venues have an advantage when the NBBO is constrained by minimum pricing requirements. While the change in minimum pricing increment provides a very clear exogenous event to test changes in dark venue market share, the behavior of dollar stocks might not be representative of the universe of stocks. To address this issue, we next study higher priced stocks and consider whether dark venues are used more actively when trading is constrained. If queue jumping is a significant competitive factor, then we expect to find a larger dark pool market share when the bid-ask spread is constrained at its penny minimum and the best bid and ask prices have sizable depth.

Due to data availability, the initial sample contains the 120 stocks used in the stratified sample analyzed by Brogaard, Hendershott, and Riordan (2013), which is chosen to include a wide range of market capitalizations and liquidity levels. Our sample period is January 3, 2011 to March 31, 2011. Three stocks are delisted before the sample period (BARE, CHTT, and KTII) and another stock is delisted during the sample period (BW), leaving a final sample of 116 stocks, which are presented in the Internet Appendix. Of these stocks, 59 are listed on the NYSE and 57 on Nasdaq. We obtain trade and quote data for this sample following the same procedures used for the dollar stock sample. Table 5 presents summary statistics for our stratified sample of stocks and shows that it contains a broad mix of spread constrained and unconstrained stocks. Compared with our dollar stock sample, we observe in our higher priced stocks larger EXCH, DARK-ECN, and BLOCK market shares, reflecting greater levels of institutional interest in these stocks.[28] INTERNALIZE market share,

[26] Under Nasdaq listing rules, stocks are subject to delisting if the bid price remains below $1.00 for 30 consecutive days. The company then has a period of 180 calendar days to re-achieve compliance (see Equity Rules Section 5810, Notification of Deficiency by the Listing Qualifications Department).

[27] Dark pools typically have lower fees than exchanges. Some midsize dark pools could have a change in fee structures for selected clients around the $1.00 threshold, but this reduction in dark pool fees is likely to be more than offset by a reduction in exchange rebates as the stock price falls below $1.00.

[28] In contrast, PING venues, also used by institutional traders, show a fall in market share, possibly due to more competition by BLOCK and DARK-ECN venues as the probability of trade execution in these venues increases for larger, more liquid stocks.

Table 5
Summary statistics: higher priced stocks.

The table reports statistics for a stratified sample of 116 stocks trading in the first three months of 2011. Panel A and B report statistics for stock and venue characteristics, respectively. *MCap* is the stock's market capitalization on January 3, 2011 in billions, and *Volatility* is the standard deviation of daily returns over the sample period. All remaining variables are measured daily. *Price* is the average trade price in dollars. *Qspread* is the time-weighted average difference between the National Best Bid and Offer prices in dollars. *Volume* is the average number of shares traded in millions. *Ntrades* is the average number of transactions in thousands. *TradeSize* is the average daily trade size in number of shares. *Mktshare* is the number of shares traded on a specific trading venue as a percentage of the total number of shares traded across all trading venues. Trades are classified into six trading venue types: exchange (EXCH), dark electronic communication networks (DARK-ECN), block crossing networks (BLOCK), ping destinations (PING), retail market makers (INTERNALIZE), and those not previously classified, (OTHER). Panel C reports statistics for *Constrain*, which is calculated as the difference between the natural log of the observed spread and the natural log of the predicted spread.

Variable	Mean	Standard deviation	Q1	Median	Q3
Panel A: Stock characteristics					
MCap (billions)	20.72	44.95	0.644	2.256	20.77
Price	46.44	70.04	16.26	30.79	54.35
Qspread	0.038	0.071	0.011	0.020	0.035
Volume (millions)	4.526	10.76	0.221	0.449	3.518
Ntrades (thousands)	17.46	28.60	1.347	3.298	22.07
TradeSize	170.6	62.77	137.5	152.4	171.6

Panel B: Venue characteristics

Trading venue	Volume (tens of thousands)		*Ntrades* (hundreds)		*TradeSize*		*Mktshare*	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
EXCH	313.2	33.92	135.7	28.33	153.4	134.0	74.29	74.47
DARK-ECN	41.53	3.592	16.35	2.25	168.3	154.5	7.845	7.760
BLOCK	5.140	1.593	0.90	0.34	618.2	516.6	2.317	1.885
PING	21.76	1.827	7.20	1.12	222.3	211.5	4.538	4.420
INTERNALIZE	40.32	2.457	7.32	0.80	334.0	299.5	5.777	4.863
OTHER	30.62	3.018	7.18	0.90	365.6	301.4	5.545	5.167

Panel C: Constrain

Variable	Mean	Standard deviation	Q1	Median	Q3
Constrain	0.371	0.457	0.077	0.316	0.576

which is made up primarily of retail trading interest, is much lower for the higher priced stocks, which is consistent with the Barber and Odean (2000) finding that households invest more heavily in small, high-beta stocks.

Dark venues are likely to be more attractive for highly spread constrained stocks. Our analysis proceeds in two stages. In the first stage, we estimate a daily measure of the spread constraint (*Constrain*) based on stock and trade characteristics. In the second stage, we relate *Constrain* to *Mktshare*. In the cross section, we expect relatively more trading in dark venues for stocks more constrained by the minimum pricing requirement. In time series, we expect relatively more dark venue trading on more spread constrained days.

We estimate a predicted spread for each sample stock using factors shown in the literature to influence spreads including price, trade activity, and return volatility (see McInish and Wood, 1992; Stoll, 2000; Bessembinder, 2003). This is formalized in the statistical model

$$Ln(Qspread_{it}) = \beta_0 + \beta_1 \text{ Ln}(Mcap_{it}) + \beta_2 \text{ Ln}(Price_{it}) + \beta_3 \text{ Ln}(Ntrades_{it}) + \beta_4 \text{ Ln}(Tradesize_{it}) + \beta_5 \, Volatility_{it} + \epsilon_i, \tag{3}$$

where $Qspread_{it}$ is the daily time-weighted quoted spread, $Mcap_{it}$ is the stock's market capitalization, and $Volatility_{it}$ is the natural log of the stock's daily high price divided by daily low price, measured over our three month sample period.[29] We estimate the equation using Tobit and OLS regressions. The Tobit regression model is bounded below at -4.6, which represents the natural log of the minimum allowed spread (0.01). The OLS regression uses all stock days that are unconstrained. Based on prior studies, we expect $Qspread_{it}$ to be positively correlated with $Price_{it}$ and $Volatility_{it}$ and negatively correlated with $Mcap_{it}$ and $Ntrades_{it}$. In untabulated results, we find that the two statistical models have good explanatory power (pseudo-R^2=0.68; R^2=0.81) and have estimates consistent with the findings of prior studies.

Next, we use the estimated coefficients and the observed daily control variable levels to predict $\text{Ln}\left(\widehat{Qspread_{it}}\right)$ for the sample of 116 stocks on each day over our sample period. Using

this predicted spread, we estimate the size of the spread constraint as[30]

$$Constrain_{it} = \begin{cases} \text{Ln}(Qspread_{it}) - \text{Ln}\left(\widehat{Qspread_{it}}\right) & \text{for Ln}(Qspread_{it}) > \text{Ln}\left(\widehat{Qspread_{it}}\right) \\ 0 & \text{Otherwise} \end{cases}. \tag{4}$$

Positive values of $Constrain_{it}$ indicate that trading in the stock is spread constrained as the observed spread exceeds its predicted value. This recognizes that stocks trading at a spread wider than a penny can also be constrained by minimum tick size requirements. For example, if the true spread is \$0.015 but the observed spread is \$0.02, trading in this stock is also considered spread constrained. In these circumstances, traders have an incentive to migrate their order flow to dark venues to benefit from the tighter pricing grids. Table 5, Panel C, reports summary statistics for *Constrain*. For an observed quoted spread of a penny, the mean *Constrain* corresponds to a predicted spread of 0.69 pennies [Ln(0.01) − Ln(0.0069) = 0.371].

We hypothesize that dark venues are more attractive to traders as stocks become more spread constrained. Table 6 estimates the relation between *Constrain* and dark venue *Mktshare* in the cross section. Given that we have only three consecutive months of trading data, there is unlikely to be much intra-stock variation. For our cross-sectional tests, we average *Mktshare* and *Constrain* over the sample period so that we have one observation per stock. Consistent with our hypothesis, we find that *Constrain* is correlated with lower *Mktshare* on exchange trading venues and higher *Mktshare* in all dark venues except for BLOCK. Panels A and B show similar results, so in the subsequent regressions we present results only based on Tobit estimates of *Constrain*.

Despite limited time series variation in the data, Table 7 estimates in time series the relation between *Constrain* and dark venue *Mktshare*. In Panel A, we establish that, for trading days when a stock is more spread constrained, dark venue market share is higher, after controlling for stock and day fixed effects. We find a higher market share in DARK-ECN, BLOCK, and OTHER venues. But, the largest difference is within DARK-ECN venues. Our regression results show an average market share of 3.0% for unconstrained stocks (i.e., *Constrain* = 0), which rises by 1.3% when the actual spread is constrained at its mean level (i.e., 0.371). Given average daily turnover of \$190 million, this yields a mean \$2.53 million (\$190 x 1.3%) shift in daily order flow from traditional exchanges to DARK-ECN venues for our sample.

In Table 7, Panels B and C, we investigate the relation between limit order book queue size and dark venue market share for spread constrained and unconstrained stocks.[31] If long queues exist in the limit order book, a trader must submit a more aggressive quote to execute sooner. Hence, strong incentives exist to shift order flow toward dark venues where trades can occur at or within the NBBO. We use the depth at the best bid and ask prices

[29] Results remain unchanged if the first-stage model is estimated from two months of data prior to the sample period.

[30] Truncating *Constrain* at zero could bias our estimates. Repeating the analysis without truncating yields similar results. In the second stage regressions, *Constrain* is an independent variable derived from the first-stage model residuals, not predictions, so adjusting second stage standard errors is unnecessary (see Pagan, 1984, Model 4).

[31] Additional robustness tests shown in the Internet Appendix divide the sample into constrained and unconstrained stocks based on the average time-weighted quoted spread. We find that the results of our analysis of higher priced stocks, presented in Tables 6 and 7, are driven by the subset of stocks that are tightly constrained by the minimum pricing increment, which is consistent with our earlier findings based on stocks crossing the \$1.00 threshold.

Table 6
Relation between *Mktshare* and *Constrain* in the cross section.

The table reports the cross-sectional regression of *Mktshare* against *Constrain* for the stocks described in Table 5. The dependent variable is *Mktshare*, the number of shares traded for the indicated trading venue type as a percentage of the total number of shares traded. *Constrain* is a daily measure of the severity of the spread constraint. We estimate *Constrain* using a Tobit regression model in Panel A and ordinary least squares (OLS) in Panel B (see text for the regression equation). For each stock, *Mktshare* and *Constrain* are averaged over all the days in our sample period. We report heteroskedasticity-robust standard errors in parentheses. ***, **, and * indicate a significance level of 1%, 5%, and 10%, respectively. EXCH = exchange; DARK-ECN = dark electronic communication network; BLOCK = block crossing networks; PING = ping destinations; INTERNALIZE = retail market makers; OTHER = others.

Variable	EXCH	DARK-ECN	BLOCK	PING	INTERNALIZE	OTHER
Panel A: Tobit regression model						
Constrain	−6.177***	1.992***	−1.539***	0.433*	3.536***	1.532***
	(0.88)	(0.34)	(0.23)	(0.23)	(0.71)	(0.47)
Constant	76.331***	7.185***	2.827***	4.395***	4.608***	5.038***
	(0.55)	(0.19)	(0.21)	(0.16)	(0.27)	(0.24)
Number of observations	116	116	116	116	116	116
Adjusted *R*-squared	0.217	0.156	0.119	0.007	0.193	0.078
Panel B: OLS regression model						
Constrain	−6.728***	2.142***	−1.667***	0.448*	3.952***	1.653***
	(1.00)	(0.39)	(0.26)	(0.25)	(0.80)	(0.54)
Constant	76.397***	7.173***	2.840***	4.398***	4.538***	5.026***
	(0.56)	(0.19)	(0.21)	(0.16)	(0.28)	(0.25)
Number of observations	116	116	116	116	116	116
Adjusted *R*-squared	0.205	0.143	0.111	0.004	0.192	0.072

on the exchange to measure queue size. According to the queue jumping hypothesis, we expect DARK-ECN market share to increase with depth. This relation is expected to be strongest for stocks with spreads severely constrained by the minimum pricing increments.

For our 116 stocks, we calculate the average value of *Constrain* over the sample period. Stocks that exhibit a *Constrain* value in the top (bottom) quartile of stocks when ranked from highest to lowest are treated as constrained (unconstrained). Consistent with the queue jumping hypothesis, we see in Table 7, Panel B, that dark trading venues (DARK-ECN, PING, and OTHER) are more attractive and EXCH venues are less attractive on days with longer order queues for constrained stocks. Thus, when spreads are tight and a large buildup of depth exists, traders take advantage of DARK-ECN venues to avoid the exchanges' long queues. In contrast, unconstrained stocks where spreads are wider and queues are relatively smaller, shown in Table 7, Panel C, exhibit is no positive relation between depth and DARK-ECN market share.[32]

Taken together, our results provide further support for the conclusion that dark pools have a competitive advantage over traditional exchanges when trading is spread constrained. Specifically, we observe a movement of order flow toward dark trading venues when trading on the exchange becomes more constrained by the minimum pricing increment, leading to significant shifts in trading venue market shares.

7. Conclusion

US stock exchanges have experienced a dramatic loss of market share in recent years, while dark trading venues have experienced a remarkable gain in market share, accounting for over a third of total shares traded in early 2013. We investigate whether this rapid growth in dark pool trading is in part driven by Securities and Exchange Commission-imposed trading rule differences across trading venues that yield significant competitive advantages. We show that differential market regulation has given dark pools an important economic advantage, leading to more dark pool trading at the expense of traditional exchanges. One clear result is increased market fragmentation of the US stock markets.

Dark trading venues compete for order flow in a variety of ways. We use regression discontinuity analysis to isolate the effects of spread constraints on competition for order flow between exchanges and dark pools. The change in minimum pricing increment from $0.01 to $0.0001 when stock prices fall below $1.00 provides a clean natural experiment to test whether spread constraints provide some trading venues with an important competitive advantage. When spreads on traditional exchanges are constrained by minimum pricing increments, traders have incentives to migrate toward dark trading venues to bypass long queues of displayed limit orders on exchanges and, thus, reduce their risk of delayed execution, while experiencing lower risk of nonexecution in the dark pool.

Consistent with the prediction that dark pools successfully attract order flow when the NBBO is constrained, we find that, when stocks trade just above $1.00, DARK-ECN market share rises to approximately double that observed when the same stock is trading just below $1.00. Further, the rise in DARK-ECN market share corresponds to a fall in traditional exchanges' market share. Importantly, these conclusions continue to hold when we examine a stratified sample of higher priced stocks that trade at the minimum

[32] Stock splits can also have a large impact on stock liquidity. For stocks priced above a dollar, reverse stock splits typically decrease the strength of a stock's spread constraint and forward splits increase the strength of the spread constraint. In results reported in the Internet Appendix, we find a large drop in dark venue market share following large reverse stock splits that reduce the effectiveness of a stock's minimum spread constraint and a rise in dark venue market share following large splits, which raises the binding nature of the minimum spread.

Table 7
Relation between *Mktshare*, *Constrain* and *Depth* in the time series.

The table reports the regression of *Mktshare* against *Constrain* for the stocks described in Table 5. The dependent variable is *Mktshare*, the number of shares traded for the indicated trading venue type as a percentage of the total number of shares traded. *Constrain* is a daily measure of the severity of the spread constraint (see text for the construction of *Constrain*). In Panel B, we regress *Mktshare* against *Log(Depth)*, where *Depth* is the daily time-weighted depth across all exchange venues at the prevailing best bid and ask prices. Panels B and C contain subsets of constrained and unconstrained stocks respectively based on the stock's average value of *Constrain*. A stock is considered constrained (unconstrained) if *Constrain* is in the top (bottom) quartile of *Constrain* across all stocks in the sample. All regressions control for stock and day fixed effects. Standard errors clustered by stock and day are in parentheses (Thompson, 2011). ***, **, and * indicate a significance level of 1%, 5%, and 10%, respectively. EXCH=exchange; DARK-ECN=dark electronic communication network; BLOCK=block crossing networks; PING=ping destinations; INTERNALIZE=retail market makers; OTHER=others.

Variable	EXCH	DARK-ECN	BLOCK	PING	INTERNALIZE	OTHER
Panel A: Baseline regression						
Constrain	−4.167***	3.595***	1.372**	−0.932***	−2.574***	2.614***
	(1.46)	(0.80)	(0.65)	(0.32)	(0.47)	(0.75)
Constant	70.723***	2.995***	−1.398*	6.962***	19.041***	1.903**
	(1.89)	(1.07)	(0.81)	(0.44)	(0.59)	(0.91)
Number of observations	7,192	7,175	6,583	7,171	7,174	7,131
Adjusted *R*-squared	0.285	0.163	0.106	0.286	0.586	0.116
Panel B: Constrained stocks sample						
Log(Depth)	−2.515***	0.719*	−0.138	0.687***	0.114	1.315***
	(0.72)	(0.38)	(0.19)	(0.22)	(0.28)	(0.35)
Log(MCap)	5.944*	−2.034	−1.781**	−0.047	1.283	−3.957**
	(3.10)	(1.46)	(0.77)	(0.91)	(1.19)	(1.82)
Log(Price)	11.967**	−3.601	−2.546*	1.146	−1.943	−4.330
	(5.30)	(2.48)	(1.49)	(1.41)	(2.02)	(2.89)
Volatility	−13.866	−11.310	4.565	−4.480	16.134**	1.675
	(29.66)	(15.26)	(4.27)	(4.97)	(6.66)	(12.70)
Constant	−80.986	57.219	50.808**	−3.745	−10.162	96.985*
	(86.47)	(40.48)	(22.80)	(24.38)	(33.11)	(50.04)
Number of observations	1,798	1,786	1,725	1,789	1,788	1,783
Adjusted *R*-squared	0.458	0.327	0.079	0.379	0.726	0.199
Panel C: Unconstrained stocks sample						
Log(Depth)	0.981	−1.037**	−0.775	−0.524***	−0.020	1.200
	(0.86)	(0.52)	(0.59)	(0.06)	(0.12)	(0.78)
Log(MCap)	−2.357	0.913	3.035	−1.285**	−0.932	1.231
	(6.47)	(2.39)	(2.89)	(0.55)	(0.92)	(2.39)
Log(Price)	6.067	−1.987	−7.259	2.157*	1.664	−2.261
	(14.32)	(5.32)	(6.39)	(1.22)	(2.05)	(5.25)
Volatility	−9.698	−4.424	−0.229	−1.443	14.581***	−3.942
	(26.34)	(10.86)	(10.32)	(3.79)	(4.30)	(14.03)
Constant	99.583	1.476	−35.598	30.779***	21.880	−24.560
	(98.44)	(35.86)	(43.47)	(8.34)	(13.87)	(37.44)
Number of observations	2,480	2,480	2,277	2,476	2,478	2,474
Adjusted *R*-squared	0.202	0.094	0.087	0.295	0.229	0.112

penny spread or that trade at spreads of two or three pennies when predicted spreads are less.

As more order flow migrates to dark pools because of the minimum tick size regulations, the probability of execution in dark pools rises, which encourages more traders to submit their orders to these trading venues. This positive feedback loop in liquidity, initially triggered by minimum pricing requirements, is another important factor driving order flow off the exchanges and encouraging the rapid growth of dark venues.

Our study shows that strong demand exists for tighter tick sizes in many highly liquid stocks. Our analysis indicates that price discovery is taking place in off-exchange venues when bid-ask spreads on traditional exchanges are tick size constrained by government regulation. Over time, the ability to queue jump on some dark venues can discourage traders from providing liquidity to traditional stock exchanges, resulting in wider spreads and less depth. Future work should consider the impact of imposing differential trading venue regulation on market fragmentation, exchange market quality, and how reduced transparency affects capital markets.

These concerns are reflected in the rapidly changing regulatory landscape for dark trading, with regulators in Australia and Canada recently requiring a minimum level of price improvement for dark trades in an effort to level the playing field between dark pools and traditional exchanges.[33] However, queue jumping and subpenny

[33] In 2012, the Investment Industry Regulatory Organization of Canada imposed a new regulation requiring dark orders to provide one full tick of price improvement relative to the prevailing NBBO or a half tick if trading is already constrained to one tick. Foley and Putnins (2013) find a sharp reduction in dark trading when Canadian regulators introduced minimum price improvement requirements for dark pools on October 15, 2012. From May 26, 2013, the Australian Securities and Investments Commission requires trades exempt from the pre-trade transparency requirement to offer meaningful price improvement of at least one price step or be at the NBBO midpoint.

pricing are only two of the ways trading venues compete for order flow. Other features, such as pre-trade opacity and exemptions from fair access requirements, also influence competitive positions of trading venues. For regulation to evolve, a deeper understanding is needed of how these trading rule differences affect intermarket competition, both individually and in combination, their impact on the competitive positions of alternative trading venues, and their ultimate effects on the quality of capital markets.

Appendix A

See Table A1.

Table A1
Variable definitions.
This table provides variable definitions.

Variable	Definition
Panel A: Main variables	
DollarPrice	An indicator variable that takes the value of one if the price bin is equal to or greater than \$1.00 and a value of zero if price bin is less than \$1.00
Constrain	The size of the spread constraint is estimated as: $Constrain_{it} = \begin{cases} \text{Ln}(Qspread_{it}) - \text{Ln}\left(\widehat{Qspread_{it}}\right) & \text{for Ln}(Qspread_{it}) > \text{Ln}\left(\widehat{Qspread_{it}}\right) \\ 0 & \text{Otherwise} \end{cases}$ where $\text{Ln}\left(\widehat{Qspread_{it}}\right)$ is predicted from the model $\text{Ln}(Qspread_{it}) = \beta_0 + \beta_1 \text{Ln}(Mcap_{it}) + \beta_2 \text{Ln}(Price_{it}) + \beta_3 \text{Ln}(Ntrades_{it}) + \beta_4 \text{Ln}(Tradesize_{it}) + \beta_5 Volatility_{it} + \epsilon_i$.
Mktshare	The number of trades for the indicated venue type divided by the total number of trades across all venue types per trading interval.
Panel B: Control variables	
MCap	For the dollar stock sample: A stock's market capitalization on day 0 in millions of dollars. Day 0 is the first day the closing price of the stock is above or below \$1.00. For the sample of stocks priced above a dollar throughout the sample period: The stock's market capitalization on January 3, 2011 in billions is used.
Ntrades	A stock's average daily number of transactions between 9:30:30 and 16:00:00 over all trading venues, or for an individual venue type, as indicated.
Price	A stock's average daily trade price between 9:30:30 and 16:00:00 over all trading venues.
Qspread	The time-weighted average difference between the National Best Bid and Offer prices between 9:30:30 and 16:00:00.
TradeSize	A stock's average daily trade size between 9:30:30 and 16:00:00 over all trading venues, or for an individual venue type, as indicated.
Volatility	The difference between the log of the intraday high price and the log of the intraday low price between 9:30:30 and 16:00:00 over all trading venues.
Volume	A stock's average daily volume between 9:30:30 and 16:00:00 over all trading venues, or for an individual venue type, as indicated.

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